Wave Polarization

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Lecture Objective

- To learn about the concept of wave polarization and how to specify the polarization state of an electromagnetic plane wave.

Polarization of Plane Waves

- The IEEE definition of polarization:

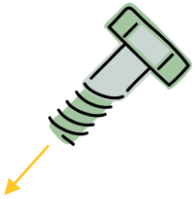
"Polarization is that property of an electromagnetic wave describing the time-varying direction and relative magnitude of the electric field vector; specifically, the figure traced as a function of time by the extremity of the vector at a fixed location in space, and the sense in which it is traced, as observed along the direction of propagation."

Polarization of Plane Waves

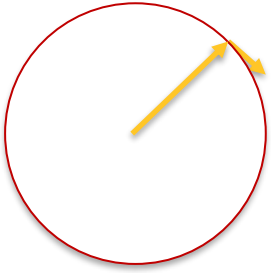
- Polarization describes the locus traced by the tip of the electric field vector as a function of time at a fixed position in space in a plane perpendicular to the direction of propagation.
- For a time-harmonic wave, the shape traced out over one period is, in general, an ellipse and this shape repeats itself over each period.
- Polarization has two properties:
  - Shape
  - Sense

An Example of Polarization in Every Day Life

- Consider a bolt. Most bolts are “right-handed” although “left-handed” bolts do exist.
- As the bolt ‘propagates’ a vector from the axis of the screw to a point on the thread in a fixed plane perpendicular to the direction of propagation rotates CW and traces out a circular shape.



Direction of propagation



Linear Polarization

- Example:

$$\tilde{\vec{E}} = \hat{x}e^{-jkz}$$

$$\vec{E}(z,t) = \hat{x}\cos(\omega t - kz)$$

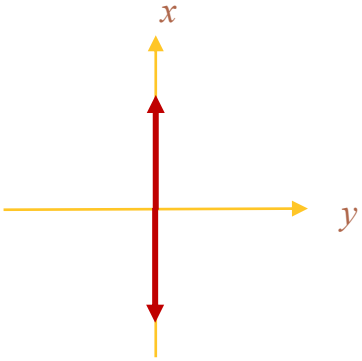
To observe polarization, pick a fixed plane perpendicular to the direction of propagation:

$$\vec{E}(0,t) = \hat{x}\cos(\omega t)$$

Linear Polarization

$$\vec{E}(0,t) = \hat{x}\cos(\omega t)$$

ωt	E_x	E_y
0	1	0
π/2	0	0
π	-1	0
3π/2	0	0
2π	1	0



Circular Polarization

- Example:

$$\tilde{\vec{E}} = (\hat{x} - j\hat{y})e^{-jkz}$$

The x- and y-components of electric field are in phase quadrature.

$$\vec{E}(z,t) = \hat{x}\cos(\omega t - kz) + \hat{y}\sin(\omega t - kz)$$

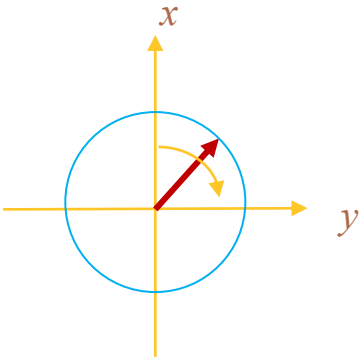
To observe polarization, pick a fixed plane perpendicular to the direction of propagation:

$$\vec{E}(0,t) = \hat{x}\cos(\omega t) + \hat{y}\sin(\omega t)$$

Circular Polarization

$$\vec{E}(0,t) = \hat{x}\cos(\omega t) + \hat{y}\sin(\omega t)$$

ωt	E_x	E_y
0	1	0
π/2	0	1
π	-1	0
3π/2	0	-1
2π	1	0



'RHCP'
'CWCP'

Circular Polarization

- Example:

$$\tilde{\vec{E}} = (\hat{x} + j\hat{y})e^{-jkz}$$

The x- and y-components of electric field are in phase quadrature.

$$\vec{E}(z,t) = \hat{x}\cos(\omega t - kz) - \hat{y}\sin(\omega t - kz)$$

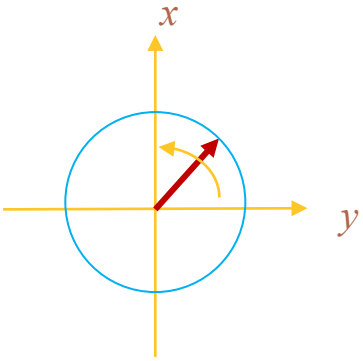
To observe polarization, pick a fixed plane perpendicular to the direction of propagation:

$$\vec{E}(0,t) = \hat{x}\cos(\omega t) - \hat{y}\sin(\omega t)$$

Circular Polarization

$$\vec{E}(0,t) = \hat{x}\cos(\omega t) - \hat{y}\sin(\omega t)$$

ωt	E_x	E_y
0	1	0
π/2	0	-1
π	-1	0
3π/2	0	1
2π	1	0



'LHCP'
'CCWCP'

Elliptical Polarization

- Lines and circles are special cases of ellipses.
- Thus, all possible polarization states of a given plane wave can be specified by the shape of the ***polarization ellipse*** (axial ratio and tilt angle) and the sense of rotation.
- A very general expression for the electric field of a plane wave propagating in the +z-direction is

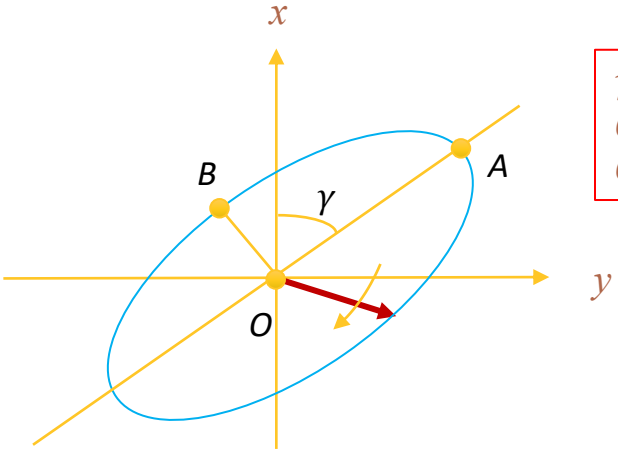
$$\begin{aligned}\vec{E}(z,t) &= \hat{x}A_x\cos(\omega t - kz + \phi_x) + \hat{y}A_y\cos\left(\omega t - kz + \phi_y\right) \\ &= \hat{x}A_x\cos(\omega t - kz + \phi_x) + \hat{y}A_y\cos(\omega t - kz + \phi_x + \delta)\end{aligned}$$

Elliptical Polarization

- Where $\delta = \phi_y - \phi_x$
- Note that it is the difference in phase between the x- and y-components that affects polarization, not the absolute phase.
- Hence, we can simplify and assume $\phi_x = 0$
- The expression for electric field becomes

$$\vec{E}(z,t) = \hat{x}A_x \cos(\omega t - kz) + \hat{y}A_y \cos(\omega t - kz + \delta)$$

Polarization Ellipse



γ is the "rotation (tilt) angle"
OA is the semi-major axis of the ellipse
OB is the semi-minor axis of the ellipse

Axial Ratio

- The **axial ratio** (*AR*) of the polarization is the ratio of the lengths of the major and minor axes in the polarization ellipse; and is often expressed in dB.

$$AR = OA/OB$$

$$AR\ (dB) = 20\log_{10}(AR)$$

Axial Ratio

- The ellipticity angle (χ) is defined by

$$\tan \chi = \pm \frac{1}{AR}$$

- By convention, the "+" sign is used for left-handed sense and the "-" for right-handed sense.
- The angle χ has the range

$$\frac{-\pi}{4} \leq \chi \leq \frac{-\pi}{4}$$

Polarization State

- The polarization state is completely specified by the rotation (tilt) angle (γ) and the (signed) ellipticity angle (χ).
- These quantities are determined from the phase difference between x- and y-components (δ) and the ratio of the magnitudes of the components.
- The latter quantity is usually specified as an auxiliary angle defined by

$$\tan \psi_0 = \frac{A_y}{A_x} \qquad 0 \leq \psi_0 \leq 90^0$$

Polarization State

- We have

$$\tan 2\gamma = (\tan 2\psi_0)\cos\delta$$
$$\sin 2\chi = (\sin 2\psi_0)\sin\delta$$

- The range of these angles are

$$\frac{-\pi}{2} \leq \gamma \leq \frac{\pi}{2}$$
$$\frac{-\pi}{4} \leq \chi \leq \frac{\pi}{4}$$

Polarization State

- Figure 7-12 shows polarization ellipses (with sense of polarization) corresponding to various polarization states defined in terms of the polarization angles (γ,χ).
- For left hand circular polarization, the value of χ is 45°. The value of γ is irrelevant.
- For right hand circular polarization, the value of χ is -45°. The value of γ is irrelevant.
- For linear polarization, the value of χ is 0°. For x-polarization that value of γ is 0°. For y-polarization that value of γ is -90° or 90°.

Polarization States for Certain Polarization Angles

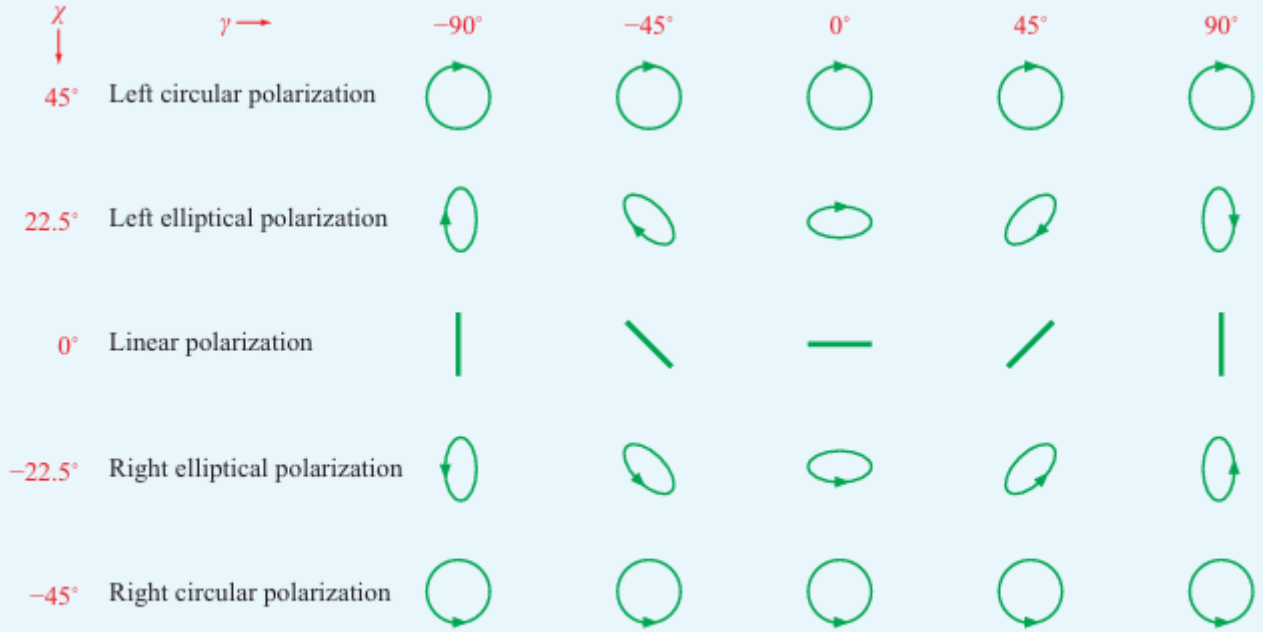


Figure 7-12 Polarization states for various combinations of the polarization angles (γ, χ) for a wave traveling out of the page.

Question

- Show that the sign of the rotation angle γ is the same as the sign of $\cos\delta$, and the sign of the ellipticity angle χ is the same as the sign of $\sin\delta$.
- Hint: Use equations (7.59) to demonstrate that the facts above are true.
- Try to offer a physical explanation for these facts.

Polarization Division Multiplexing of Signals

- If we think of all possible polarization states of a plane wave as a linear space, that linear space has a dimensionality of 2.
- A consequence of this fact is that we can receive two independent radio signals that are otherwise identical (in frequency, direction of arrival, location of source, etc.) by sending them on two orthogonal polarizations (e.g., vertical/horizontal, or RHCP/LHCP).
- Polarization diversity is heavily exploited in satellite systems such as DBS-TV.

Summary

- Polarization is an important property of an electromagnetic wave.
- The polarization state is specified by the two polarization angles.

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